

ChronoPDE V2

A sealed five-seed comparison of matched cosine and Fourier neural operators

Confirmatory outcome

The predeclared Phase 6 gate passed. Across five frozen checkpoint pairs and 256 sealed trajectories, the DCT operator had lower rollout and exact-velocity error in every seed.

Measure	Frozen result
Median paired rollout improvement	21.14%
Hierarchical-bootstrap 95% interval	16.35%-27.72%
Directional wins	5/5 (one-sided p=0.03125)
DCT divergence	0 / 1,280 rollouts
FFT divergence	255 / 1,280 rollouts

Claim boundary: this result applies to the frozen reaction-diffusion PDE, 64 x 64 grid, central parameter range and training protocol. It is not a claim of boundary enforcement, wall-flux correctness, OOD generalization or universal DCT superiority.

1. From a negative result to a fresh study

The original V1 study used spline-derived velocity targets and unequal-history continuous-time checkpoints. Its unchanged Week 6 gate did not pass. That negative result remains preserved under the v0.1 tags.

V2 did not retune on the exposed V1 test set. It introduced exact discrete-RHS supervision, capacity matching, fresh development identities, five training seeds and a new confirmatory split whose states were generated only after all ten checkpoints and evaluation settings were frozen.

This separation makes the positive V2 outcome additive rather than a rewrite of the historical result.

2. Numerical ground truth

The supervised target is the exact finite-volume simulator RHS per unit physical time. Phase 2 checked the cell-centred Neumann Laplacian for symmetry, negative semidefiniteness, constant nullspace, diffusion mass conservation and the discrete Green identity.

Sparse-matrix, reflected-stencil and DCT-spectral operators agreed within frozen float64 tolerances. Float32 storage and a tight DOP853 convergence repeat also passed. Phase 3 generated 512 training and 128 validation trajectories; the normalizer used training data only.

The DCT basis is aligned with the discrete Neumann operator. The full network is not boundary enforcing because pointwise paths, nonlinearities, FiLM, lifting and projection can alter boundary values.

3. Matched operator comparison

Both vector fields use width 29, four blocks, the same FiLM conditioner, pointwise paths, lifting, projection, initialization and RK4 rollout. Each has 1,973,657 real trainable parameters and 484,416 active real spectral degrees of freedom per block.

The FFT control retains 12 x 12 complex Fourier modes. ChronoPDE retains 24 x 24 real orthonormal cosine modes. The 4.35% physical-cutoff mismatch is within the predeclared 10% tolerance.

Consequently, the comparison isolates the basis-aligned spatial inductive bias far more tightly than the V1 models.

Component	FFT control	ChronoPDE DCT
Retained modes	12 x 12	24 x 24
Parameters	1,973,657	1,973,657
Real spectral DOF/block	484,416	484,416
Spatial assumption	periodic	Neumann-aligned

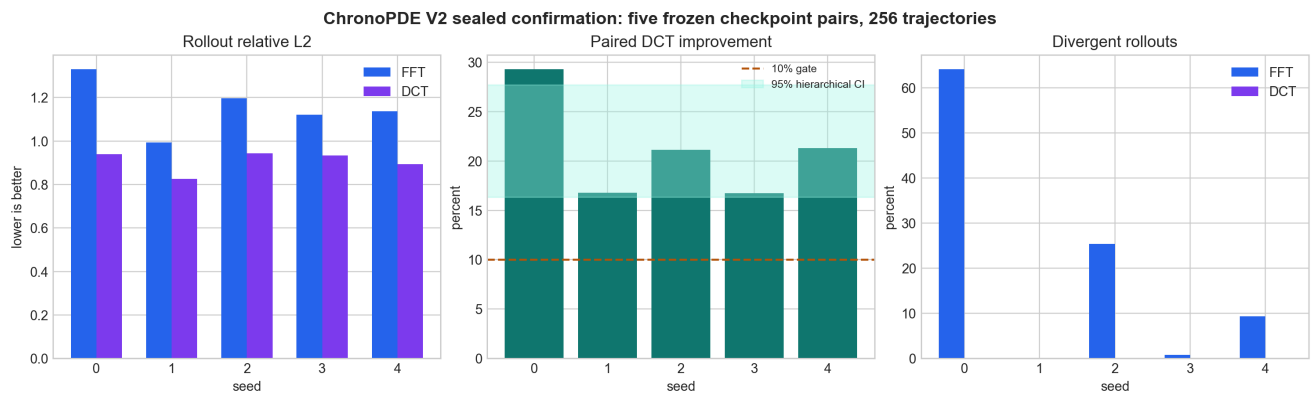
4. Development and sealed protocol

Seeds 0-4 were trained for each backbone with identical exact-RHS objective, optimizer, budget and development data. Checkpoints were selected only by fixed validation velocity nRMSE, with validation relative loss as tie-breaker.

The 256 confirmatory identities were frozen in Phase 2 and remained ungenerated through model development. Phase 6 generated them once, verified 32 exact-RHS samples and 12 tight-solver repeats, then evaluated all frozen checkpoints once.

Finite predictions above magnitude 10 remained in rollout errors and were also marked divergent. Nonfinite predictions were unavailable and treated as infinite for the architecture decision.

5. Confirmatory results



Seed	FFT rollout	DCT rollout	DCT improvement	FFT div.	DCT div.
0	1.3300	0.9403	29.30%	64.06%	0.00%
1	0.9927	0.8259	16.80%	0.00%	0.00%
2	1.1966	0.9437	21.14%	25.39%	0.00%
3	1.1212	0.9336	16.73%	0.78%	0.00%
4	1.1364	0.8941	21.32%	9.38%	0.00%

Every seed cleared the directional rollout and velocity requirements. The median paired improvement exceeded the 10% practical-effect gate, and the bootstrap interval remained wholly positive.

6. Stability, boundary attribution and provenance

DCT produced zero divergent rollouts. FFT divergence was concentrated in seed 0 but appeared in four of five seeds overall. Phase 4B had already shown that the seed-0 FFT instability persisted under RK4 refinement, supporting a learned-vector-field explanation rather than an integration-resolution artifact.

DCT also had lower boundary-strip and first-interior normal-derivative error in all five seeds. This authorizes the phrase lower boundary-region error only; the first-interior metric is not physical wall flux.

Frozen provenance

- Confirmatory HDF5 SHA-256:
0a432d24d34605274a4e4bfa06d97831602062b016b44f4d5b81f91cc14e64e3
- Phase 6 results ZIP SHA-256:
151e0fb628735b21f7d63a159abfa9695253f2a210147c29555ced06d587cfc1
- Protocol SHA-256: 7a869166876c6bd2a6143bd334a71128f692b7dc77cc9554d69e510b09c26464
- Decision evidence: reports/chronopde_v2/phase6/decision_report.json
- Paired seed evidence: reports/chronopde_v2/phase6/paired_seed_results.csv

Limitations

The study covers one PDE family, grid, parameter range, objective, optimizer, five fixed seeds and one rollout protocol. Sparse time, OOD, grid transfer and other PDEs require new registered studies and new holdouts. The Phase 6 set must not be reused for retuning.